

Zhandos Brown

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EDUCATION

Boston University

Boston, MA

B.S. Applied Mathematics

Graduation Date: 2026

- Relevant Coursework: Robotics & Autonomous Systems, Software Engineering, Data Science, Probability, Linear Algebra

WORK EXPERIENCE

Software Engineer Co-op, Mobile

Boston, MA (Remote)

BumpAutos Inc. (CarPot)

Jan 2026 - Mar 2026

- Delivered 30+ React Native screens by converting product requirements and Figma designs into reusable components
- Architected typed GraphQL data flows with Code Generator, TanStack Query, Redux Toolkit, and secure authentication, improving reliability across a 600+ file TypeScript codebase

Mobile Software Engineer Intern

Merrimack, NH

Fidelity Investments, NetBenefits Health

Jun 2025 - Aug 2025

- Reduced document-rendering latency 70%, from 300 ms to 90 ms across 100K+ daily sessions, by profiling bottlenecks and optimizing production SwiftUI components
- Improved offline reliability by migrating GraphQL document APIs to REST across Swift, Java, Node.js, and SQL systems

Research Software Engineer

Boston, MA

Boston University College of Engineering

Jun 2023 - May 2025

- Engineered ROS 2 three-drone perception system on NVIDIA Jetson that processed 500GB+ of daily multi-sensor data at sub-50 ms latency, achieving sub-2 mm localization through C++ point-cloud and SLAM pipeline with PCL and Open3D
- Achieved 94% road-defect detection accuracy across 50K+ image frames and 64 km by developing PyTorch/YOLO perception pipeline and validating model outputs against annotated field data

PROJECTS

Kinetic Robotics Studio

zhandolia.github.io/kinetic-robotics-studio

Robotics Simulation & Sensor Data Visualization

- Built and deployed a React/Three.js workspace synchronizing two calibrated camera views with 4,488 3D points per frame, camera-LiDAR overlays, playback, and versioned import/export
- Validated TypeScript/Python scene formats with 42 shared fixtures and verified multi-view geometry through 384 cuboid-corner projections and 48 image-pixel checks

Ring Dynamics

zhandolia.github.io/ring-dynamics

Boxing Biomechanics and Video-Analysis Platform

- Built a perception-driven boxing analytics platform using 200+ hours of professionally annotated footage, integrating C++/Python inference with a Next.js/FastAPI dashboard to track strikes, calculate 23 metrics at 60 FPS, and compare fighter performance

TECHNICAL SKILLS

Programming: TypeScript, JavaScript, Python, SQL, C++, Java, Swift

Graphics: React, Next.js, Three.js, WebGL, React Native, SwiftUI

Backend: FastAPI, Node.js, REST APIs, GraphQL, PostgreSQL, NumPy, Pydantic, Zod

Robotics: ROS 2, PCL, Open3D, Linux, Git, Docker, AWS, Automated Testing, CI/CD